

Effects of lower limb prosthesis on activity, participation, and quality of life: a systematic review

Prosthetics and Orthotics International
36(2) 145–158
© The International Society for
Prosthetics and Orthotics 2012
Reprints and permission:
sagepub.co.uk/journalsPermissions.nav
DOI: 10.1177/0309364611432794
poi.sagepub.com



Kersti AM Samuelsson¹, Outi Töytäri², Anna-Liisa Salminen³
and Åse Brandt⁴

Abstract

Background: Effects presented on the use of assistive devices such as prosthesis are often based on laboratory findings (i.e. efficacy).

Objectives: To summarise and evaluate findings from studies on effectiveness of lower limb prostheses for adults in real life contexts, primarily in terms of activity, participation, and quality of life (QoL) and secondarily in terms of user satisfaction, use/non-use, and/or cost-effectiveness.

Study Design: Systematic review.

Methods: We included controlled studies and non-controlled follow-up studies including both baseline and follow-up data. Using 14 different databases supplemented with manual searches, we searched for studies published from 1998 until June 2009.

Results: Out of an initial 818 identified publications, eight met the inclusion criteria. Four studies reported on the effectiveness of a microprocessor-controlled knee (MP-knee) compared to a non-microprocessor-controlled knee (NMP-knee). Results were inconsistent except for quality of life and use/non-use, where the authors reported an improvement with the MP-knee compared to the NMP-knee. The remaining four studies included a diversity of prosthetic intervention measures and types of endpoints.

Conclusions: Overall, there was an inconsistency in results and study quality. This review highlights the need for high-quality research studies that reflect the effectiveness of different prosthesis interventions in terms of users' daily living and QoL.

Clinical relevance

Clinical guidelines are important to every practitioner. Information on expected effectiveness from assistive devices should be well founded and contain both facts about the device quality and its contribution to users' daily lives. Thus, studies based on users' experiences from prosthetic use in everyday life activities are of great importance.

Keywords

Amputees, daily living, evidence

Date received: 6 January 2011; accepted: 18 November 2011

Background

Affecting many daily activities, amputation is a life-altering event with an immediate and obvious effect on a person's mobility, including many daily life activities. During the last decades, new metals and plastics, as well as computer-aided design and new developments in microprocessors, have significantly affected the evolution of lower limb prosthetics, making it possible for amputees to be more independent and physically active.^{1,2} Nevertheless, earlier

¹University Hospital, Department of Rehabilitation Medicine, Linköping, Sweden

²National Institute for Health and Welfare, Helsinki, Finland

³The Social Insurance Institution, Helsinki, Finland

⁴Danish Centre for Assistive Technology, Aarhus C, Denmark

Corresponding author:

Kersti AM Samuelsson, University Hospital, Department of Rehabilitation Medicine, SE - 581 85, Linköping, Sweden
Email: kersti.samuelsson@lio.se

studies have reported that a majority of amputees do not consider prosthetic devices useful because of pain, discomfort or misfit.^{3,4} Karmarkar et al.⁵ reported that 31% in a group of veterans with lower-limb prosthesis used wheel-chairs as their primary mobility device.

The scientific literature has paid much attention in describing new technical solutions to optimise prosthesis function, e.g. microprocessor-controlled prosthesis, new orthopaedic surgery methods, such as the osseointegrated prosthesis,^{6,7} new socket solutions, and custom-designed solutions.¹ However, there seems to be a lack of knowledge in how a prosthesis system affects a person's daily life in the user's natural context; consequently, there is a need to gather complementary evidence on device effectiveness to enrich best clinical practice.⁸ Clinicians and policymakers often distinguish between the efficacy and the effectiveness of an intervention. Efficacy trials determine whether an intervention produces the expected result under ideal circumstances while effectiveness trials measure the degree of beneficial effect under 'real world' settings.⁹

Thus, the primary purpose of this review is to summarise the current state of knowledge about how lower limb prosthesis for adults can affect activity, participation, and quality of life (QoL) in real life contexts. The secondary purpose of this review is to summarise the state of knowledge concerning user satisfaction, use/non-use, and/or cost-effectiveness when investigated together with primary outcomes.

Methods

This study is a systematic review. Controlled studies and non-controlled follow-up studies, provided that the latter includes both baseline and follow-up data, were included. The focus was on effectiveness studies, thus results collected in patients' natural contexts. Data from laboratory settings (efficacy studies) or controlled studies, including a population without amputation as controls, were excluded. The study population included lower limb prosthetic users over 18 years of age according to ISO 9999 classification¹⁰: 062403, partial foot prosthesis; 062406, ankle disarticulation prosthesis; 062409, transtibial (below-knee) prosthesis; 062412, knee disarticulation prostheses; 062415, transfemoral (above-knee) prostheses; or 062418, hip articulation prostheses. The control intervention should include some other prosthesis system or no prosthesis system.

We considered studies focusing primarily on activity and/or participation and/or QoL outcomes. We also considered studies on user satisfaction, use/non-use, and costs outcomes, which we, however, considered to be secondary. That is, we did not include studies that solely reported outcomes based on these dimensions.

Search strategy

The authors selected keywords and synonyms based on the inclusion criteria in collaboration with a research librarian. The search was performed without language restrictions in

the following databases: DARE, Cochrane, CIRRIE, Campbell, OVID MEDLINE, ASSIA, PEDro, OTDatabase, Cinahl, OT seeker, Embase, PsychINFO, Agelina, and Social Abstracts. The OVID MEDLINE search strategy was based on the following terms:

Leg prostheses/lower limb prosthetic system/prostheses/partial foot prosthesis/ankle disarticulation prostheses/transstibial prostheses/knee disarticulation prostheses/transfemoral prostheses/hip articulation prostheses/Amputation/anomalia/Outcome/effect/cost/cost-effectiveness/Use/non-use/Mobility/activity/occupation/participation/user satisfaction/quality of life/transportation/walking/moving around/running/sports/ADL/PADL/IADL/Assessment/outcome assessment/evaluation/measurement/score/scale/test/Randomized controlled trial, random/cohort studies, follow-up studies, pre-post studies.

The strategy was modified for the other database searches. A manual search of AAATE conference proceedings, RESNA proceedings, and the *Assistive Technology and Disability* journal completed the searches in different databases. Original studies, as well as reviews, were searched to find possible secondary sources. The search covered January 1998 until June 2009.

Study selection

Two authors (KS and OT) independently went through titles and abstracts of the identified studies and excluded duplicates and studies that obviously were not in accordance with the inclusion criteria. Then, potential studies identified were ordered and read in full text. Next, the same two reviewers independently suggested final inclusion or exclusion for further analysis. In case of disagreement, consensus was sought through discussion or by consulting the other two reviewers. The data extraction followed a defined and formerly used structure,^{11,12} including a description of the design, methods (Table 1), used instruments (Table 2) and results (Table 3).

Assessment of study quality

The finally included studies were read, and all four authors assessed study quality. The internal validity of the studies was assessed by a modified criteria list adopted from Borghouts et al.¹³ The assessment included a yes/no option to describe the following: a sufficient description of the population selection; inclusion and exclusion criteria described; sufficient study size (> 10 patient years); follow-up > four months; dropouts < 20%; dropouts described; outcome measures and data presentation congruent with the study aims; confounder adjusted in the analysis; and psychometric properties of the instruments reported. The modification of the original assessment schedule meant that the two latter criteria replaced the original criterion: appropriate analysis technique. This modified version has been tested in another recently published review.¹¹ Criteria for

Table 1. Study design, follow-up time, interventions, participants, inclusion/exclusion criteria, and selection in the included studies.

Reference author, year	Study design	Follow-up time	Intervention and control intervention	Participants	Inclusion/Exclusion criteria	Selection of included subjects
Coleman et al. 2004 ¹⁶	Randomised crossover trial	3 months for both intervention and control intervention.	Intervention: An elastic suspension liner with distal locking pin (Alpha [®] system) Control: A liner with neoprene suspension sleeve (Pe-Lite [™])	n = 14, dropouts = 1 Final group n = 13 Unilateral TTA of traumatic origin. Male = 77% Mean age 49.4 years (range 31.5–65.8)	Incl: Unilateral TTA of traumatic origin, > one year post amputation, stable in their current prosthesis, minimum ambulatory function of level 2 based on DMERC levels. Excl: Major pre-existing health problems.	NR
Donovan-Hall et al. 2002 ¹⁹	Prospective controlled before and after	12 weeks	Intervention: Prosthesis with a silicone cover Control: Amputees' own non-silicon prosthesis	n = 11, dropouts = 1. Final group n = 10, TTA Male = 9%. Mean age 37.4 years (range 19–59)	Incl: Age 18–60 years, lower limb amputees. Excl: Life-threatening illness	All TTA-subjects consecutively attending a limb centre for first fitting of a prosthesis, over a 6-month period.
Gerzeli et al. 2009 ²²	Cross-sectional case-control 2 groups; MP-knee NMP-knee	NA	Intervention: MP-knee Control: NMP-knee	MP group: n = 50. Unilateral TFA. Male = 88%. Mean age 45.8 ± 11.8 years NMP group: n = 50 Male = 88% Mean age 45 ± 12 years n = 21, drop outs = 4, Final group n = 17 Unilateral TFA, Male = 76%, Mean age 49 years (range 21–77)	Incl: MP group: Age 18–65 years Fitted with prosthesis > 3 years ago NMP group: Matched according to sex, age, and cause of amputation.	First, all subjects were randomly selected from the records of the INAIL centre of Bologna. Second, case-matching was performed between groups.
Hafner et al. 2007 ²⁰	Controlled reversal (A-B-A-B) A = Mechanical control prosthesis knee (NMP-knee) B = Microprocessor control prosthesis knee (MP-knee)	A1: 8 weeks B1: 8 weeks A2: 2 weeks B2: 8 weeks	Intervention: MP-knee Control: Own NMP-knee		Incl: Age ≥ 18 years, TFA unilateral, Medicare Functional Classification Level (MFCL) 2 or 3, > 2 years post amp, current use of a NMP-knee, able to ambulate without personal assistance on level ground, inclines, stairs, and uneven terrain with a NMP-knee and test socket equivalent to existing socket. Excl: Chronic residual limb skin breakdown, secondary health problems	Subjects recruited from the local amputee population.
Hagberg et al. 2008 ⁶	Prospective study with a before and after design	2 years	Intervention: Osseointegrated trans-femoral amputation prostheses (OI-prostheses) Control: Amputees' own conventional socket prosthesis	n = 18, dropouts = 1 Final group: n = 17 Unilateral TFA n = 16, Bilateral TFA n = 2. Male = 44% Mean age 45 years (range 22–62)	Incl: Age < 70 years, TFA with problems using a conventional socket prosthesis, completed maturation of skeleton and normal skeletal anatomy, suitable for surgery. Excl: Severe peripheral vascular disease, treatment with specific drugs, excessive body weight (≈ 100 kg) and pregnancy	The first 18 individuals in the Swedish OPRA study treated with surgery at the Sahlgrenska University Hospital in Göteborg during May 1999 to October 2003.

(continued)

Table 1. (continued)

Reference author, year	Study design	Follow-up time	Intervention and control intervention	Participants	Inclusion/Exclusion criteria	Selection of included subjects
Kahle et al. 2008 ¹⁸	Before- and after (A-B) A = NMP-knee B = MP-knee	A and B: 90 days	Intervention: MP-knee Control: NMP-knee	n = 21, dropouts = 2 Final group: n = 19 TFA Male = NR Mean age 51 years, (range 22–83)	Incl: TFA or knee-disarticulation amputation, use of existing NMP knee mechanism for > 90 days before study, ≥ 1 episode of physical therapy intervention, ability to ambulate in the community with or without assistive device and no human support, ability to descend stairs with no human support and willingness to continue NMP knee use for 90 days followed by MP knee use for 90 days. At least K2 according to Medicare Functional Classification Level (MFCL), be willing to comply with follow-up visits and medically cleared for ambulation with a prosthesis.	NR
Klute et al. 2006 ¹⁷	Crossover with repeated measures. Two research groups: TTA - Transfemoral amputees TFA - Transfemoral amputees	TTA group: 3 weeks acclimatization period + 1 data collection period TFA group: 3 months acclimatization period + 1 week data collection period	TTA group Intervention: Shock-absorbing pylon Control: Rigid pylon TFA group Intervention: MP-knee Control: NMP-knee	TTA group: n = 15, dropouts = 3, Final group n = 12 Male: NR Mean age 54 ± 6 years TFA group: n = 18, dropouts = 13, Final group n = 5 Male = NR Mean age 48 ± 12 years	TTA group Incl: Age 18–70 years, > 2 years prosthesis use, > 8 hours/day, walk without upper extremity aids, moderately active community ambulatory, < 120 kg, no falls within 6 months. Excl: Cause of amp a tumour, significant lower extremity pain or other neurological conditions interfering with ambulation. TFA-group Incl: Age 30–65 years, wear prosthesis > 8 hours/day, walk without upper extremity aids, walk up and down 3 flights of stairs, previous long-term NM-prosthetic knee user. Excl: Significant neurological deficit or underlying musculoskeletal disorder interfering with ambulation.	TTA subjects were recruited from VAPSHCS and Seattle, WA, metropolitan area prosthetics clinics. TFA subjects were recruited from VAPSHCS clinic.

(continued)

Table 1. (continued)

Reference author, year	Study design	Follow-up time	Intervention and control intervention	Participants	Inclusion/Exclusion criteria	Selection of included subjects
Selles et al. 2005 ²¹	Prospective randomised controlled trial	3 months	Intervention group: Total surface-bearing (TSB) socket Control group: Conventional patellar tendon-bearing (PTB) socket	$n = 36$ Intervention group: $n = 17$, dropouts = 5 Final group $n = 12$, TTA Male = NR Mean age 67.6 years Control group: $n = 19$, dropouts = 5, Final group $n = 14$ Male = NR Mean age 57.9 years	Incl: Unilateral TTA, >18 years, prosthesis user for more than 1 year, active walkers with or without a walking aid, able to bear distal end pressure. Excl: Present stump problems, problems using silicone liners.	Subjects were recruited during their visit to a rehabilitation physician or by telephone, from two hospitals in Rotterdam, between September 2000 and February 2001.

DMERC, Durable Medical Equipment Regional Carrier; Excl, Exclusion criteria; ICES TBS, Total surface-bearing socket; Incl, Inclusion criteria; MFCL, Medicare Functional Classification Level; MP, Microprocessor control prosthesis; NA, Not applicable; NAIL, Italian Workers' Compensation Authority; NMP, Mechanical control prosthesis; NR, Not reported; OPRA, Osseointegrated Prostheses for the Rehabilitation of Amputees; PTB, Patellar tendon-bearing socket; TFA, Trans-femoral amputees; TTA, Transtibial amputees; VAPSHCS, Veterans Affairs Puget Sound Health Care System.

Table 2. Outcome evaluation instruments used.

Instrument abbreviation	Instrument full name	Objective/domain(s)	Scale
AMP	Amputee Mobility Predictor	Functional capabilities of amputee patients.	Score 0–47 21 items, each score 0–1 or 0–2
Discomfort-EEARB	Discomfort related to Engaging in Everyday Activities involving Revealing the Body (EEARB) scale	How comfortable respondents think they would feel in activities of the EEARB.	Score 11–44 11-item scale. Score 1–4; 1 = very uncomfortable, 4 = totally comfortable
EEARB	Engagement in Everyday Activities involving Revealing the Body scale	How many times in the last 3 months respondents engaged in different activities which involved revealing their body.	Score 0–60 10-item scale. Score 0–6; 0 = not applicable, 6 = daily
EQ-5D	EuroQoL 5D EQ-VAS	A) Health related Quality of life, including 5 dimensions: mobility, self-care, usual activities, pain/discomfort, anxiety/depression and self estimated health. B) Self-rated health condition EQ-VAS	A) Score 0–1.0 1.0 = maximum good health B) 0–100, 100 = best imaginable health
HAD	Hospital Anxiety and Depression Scale	Anxiety and depression in people with physical disorders.	Scale 0–21 3 levels, 0–7 = normal range, 8–10 = borderline range and 11–21 = clinical range
HAI	Hill Assessment Index	Measure subjects' ability to ambulate a downgrade hill at self-selected speed.	Score 0–11 0 = cannot do/refuses to do, 11 = even step, without assistive device
PEQ	Prosthesis Evaluation Questionnaire	Self-assessed performance: ambulation, perceived response, sounds, appearance, residual limb, utility, frustration, social burden and well-being. Self-assessed satisfaction	Each scale 0–100 (visual analogue scale (VAS)) High scores = positive result Performance: 54 questions in nine performance scales. Satisfaction: one question
PEQ-A	Prosthesis Evaluation Questionnaire (PEQ) with additional questions	Additional 14 items added to the PEQ regarding subject confidence, concentration, stumbles and falls.	Each scale 0–100 (VAS) Higher scores = positive result. 14 questions
Q-TFA	Questionnaire for person with a Trans-femoral Amputation	Condition specific about: current prosthetic use score, mobility score, problem score, and global score.	Each scale 0–100 Higher score on the 'problem score' = more problems and a reduction of quality of life. Higher scores on the other scales = positive result.
RSE	Rosenberg Self-Esteem Scale	Self-esteem or self worth, which is part of wider construct of self-concept.	Score 10–40 Lower scores = higher self-esteem
SAI	Stair Assessment Index	Stair ascent and descent ability.	Score 0–13 0 = cannot do/refuses to do, 13 = without rail or assistive device, step-over-step pattern
SAM	Step-Watch™ activity monitor	Pager-sized instrument that record ambulatory activity as the number of steps taken.	
SCS	Socket Fit Comfort Score	Comfort of socket fit.	Score 0–10 0 = most uncomfortable, 10 = completely comfortable One question
SF-36	Medical Outcomes Study 36-Items Short-Form Health Survey	Health status, eight components. Role limitations due to physical health: physical functioning, role-physical, bodily pain and general health. Role limitations due to mental health: vitality, social functioning, role functioning -emotional and mental health.	Eight subscales transformed into 0–100 for each scale. 100 = best possible health

Table 3. Effectiveness of lower limb prosthesis.

Reference	Activity participation	Quality of life	User satisfaction	Use/non-use	Costs
Coleman et al. 2004 ¹⁶	No difference for in stride length between the Alpha [®] and Pe-Lite [™] liner (SAM). Subjects achieved 83% more steps/day with the Pe-Lite [™] limb compared to Alpha [®] liner (SAM). Time spend on low, moderate and high step activities/day increased 69%/77%/118% with the Pe-Lite [™] limb compared to Alpha [®] (SAM).	NR	No difference in satisfaction with prosthesis-specific issues (PEQ and SCS) between the Alpha [®] and Pe-Lite [™] . NR	Subjects spent 82% more time (SAM) wearing the Pe-Lite limb compared to the Alpha [®] .	NR
Donovan-Hall et al. 2002 ¹⁹	Using prosthesis with silicone cover showed an increase in engagement in activities involving revealing the body EEARB scores $p < 0.05$ and Discomfort-EEARB scores $p < 0.01$, compared to non-silicon prosthesis.	No change in self-esteem, anxiety, and depression (HAD) with the silicon cover	NR	NR	NR
Gerzeli et al. 2009 ²²	NA	MP knee group showed a higher level of QoL (EQ-5D) compared to the NMP knee group $p = 0.007$.	NA	NA	The cost (healthcare perspective) to gain one QALY by the adoption of the MP-knee was €35,971, which should be considered as cost-effective. If non-healthcare costs and productivity losses are included, the costs were similar for both interventions.
Hafner, et al. 2007 ²⁰	Stair descend increased (SAI) for the Microprocessor control prosthesis knee (MP) compared to amputees' own mechanical control prosthesis knee (NMP), $p < 0.001$, time to descend hill decreased with MP, $p < 0.005$ and affected-side step length increased with MP $p < 0.01$ Frequency of stumbles decreased with MP $p < 0.05$ as well as, semi-controlled falls $p < 0.05$, frequency of uncontrolled falls $p < 0.05$ and difficulty in multitasking $p < 0.05$. No difference between MP compared to NMP on: mean step frequency, mean estimated daily distance, basic mobility (AMP), indoor walking on level ground, sound side step length, affected side step length (Step-Watch 2 step activity monitor), stair ascent (SAI), the ability to descend a decline (HA), time to ambulate obstacle course or self-selected walking speed on obstacle course.	Frustration with falling (PEQ-A) decreased $p < 0.01$ with MP compared to NMP. No difference in self-assessed general health scores (SF-36).	Satisfaction (PEQ) with MP was higher than for NMP $p < 0.001$. 14 subjects preferred MP, 1 preferred NMP and 2 subjects had no preference.	Of all steps taken during two weeks, 94.4% were taken on the MP	NR

(continued)

Table 3. (continued)

Reference	Activity participation	Quality of life	User satisfaction	Use/non-use	Costs
Hagberg et al. 2008 ⁶	Mobility (Q-TFH) improved with Ol-prosthesis $p = 0.001$ compared to amputees' own conventional socket prosthesis.	QoL (SF-36) improved with Ol-prosthesis: Physical functioning $p = 0.001$, role physical functioning $p = 0.003$, bodily pain $p = 0.029$ and physical component score $p = 0.001$. The remaining five dimensions did not change. Problem score (Q-TFA) improved with Ol-prosthesis $p = 0.002$ and global score = the overall situation as an amputee $p = 0.002$.	NR	Use (Q-TFA) was improved with Ol-prosthesis $p = 0.013$ compared to amputees' own conventional socket prosthesis	NR
Kahle et al. 2008 ¹⁸	Stumbles $p = 0.006$ and falls $p = 0.03$ decreased with MP-knee. Walking speed under several conditions (SSWS 75 m, FPWS 75 m, FPWS 38 Uneven Terrain and FPWS 6 m) increased ($p < 0.01$) with MP-knee. Stair descent increased for 12 subjects (63%) with MP-knee, 2 subjects with NMP and 5 subjects showed no change.	An improvement of 20% in prosthesis-related QoL (PEQ) was shown for the MP-knee compared with the NMP-knee, $p = 0.007$.	74% wanted to keep the MP-knee and 26% wanted to keep the NMP-knee	NR	NR
Klute et al. 2006 ¹⁷	No difference in the level and duration of step activity/day (StepWatch 2 activity monitor) between the shock-absorbing pylons compared to rigid pylon in the TTA group. No difference in level and duration of activity (Step-Watch 2 activity monitor) with MP knee compared to NMP knee in the TFA group	NR	NR	NR	NR
Selles et al. 2005 ²¹	No difference in mobility-related activities and gait characteristics (Activity Monitor) between total surface bearing (TSB) socket group and conventional patellar tendon-bearing (PTB) socket group.	NR	No difference in user satisfaction (PEQ) between TSB group and PTB groups.	NR	Cost of materials was significantly higher, manufacturing time was significantly shorter, and number of visits was significantly less in the TSB group than in PTB group.

*For abbreviations of used instruments and scales, see Table 3. Discomfort-EARB, Discomfort related to Engaging in Everyday Activities involving Revealing the Body scale; EARB, Engagement in Everyday Activities involving Revealing the Body scale; EQ-5D; FPWS, Fastest possible walking speed; HAD; Hospital Anxiety and Depression Scale; HAI, Hill Assessment Index; HRQL, Health Related Quality of Life; MRPP PCS; Montreal Rehabilitation Performance Profile Composite Score; MP, Microprocessor knee; NMP, Mechanical control knee; NR, Not reported; Ol-prostheses, Osseointegrated trans-femoral amputation prostheses; PEQ, Prosthesis Evaluation Questionnaire; PEQ-A, Prosthesis Evaluation Questionnaire with additional questions; QALY, Quality adjusted life years; Q-TFA, Condition specific outcome measure that reflects current prosthetic use, mobility, problems and global health; SAI, Stair Assessment Index; SAM, Step-Watch™ activity monitor; SCS, Socket Fit Comfort Score; SF-36, Medical Outcomes Study 36-Items Short-Form Health; Survey SWS, Self-selected walking speed.

study-size, follow-up, and dropouts were applied (Table 4). The yes/no was transferred into 1/0, making it possible to summarise a score representing internal validity of the included studies (internal validity max score = 10 points).

The external validity and applicability was assessed using a modified version of a method described by Shekelle et al.¹⁴ From the original criteria, we replaced one question – treatment benefits in relation to adverse effects – with another question that we considered to be more relevant to the study – effect gain of >10% – adapted from a study on research related to back pain and functionality.¹⁴ Thus, the assessment included a yes/no transferred into 1/0 (maximum score = 4 points) to describe the following: study participants and interventions described in detail; clinically relevant outcomes measured and reported; and size of the effect clinically important (Table 5).

Furthermore, the quality assessment of the randomised controlled trial (RCT) was based on a modified version of Van Tulder et al.¹⁵ From the original 11 questions, we chose the seven aspects considered most relevant for our purpose (Table 6): the method of randomization was adequate; the treatment allocation was concealed; the groups were similar at baseline; co-interventions were avoided or similar; the compliance was acceptable; the timing of the outcome assessment was similar; and whether the analysis included an intention-to-treat analysis.

Results

The initial search resulted in 818 titles (Figure 1). After going through the abstracts, 737 studies were excluded due to not fulfilling the inclusion criteria or being duplicates. The remaining 81 studies were read in full text and another 73 studies were excluded due to not fulfilling the inclusion criteria. The remaining eight studies were controlled studies, all including outcomes of a lower limb prosthetic intervention in terms of activity and/or participation and/or QoL.

Study design and size

All studies were published within the last nine years. Of the eight included studies, seven were experimental^{6,16–21} and one was an observational study²² (Table 1). The only RCT-study was the one presented by Selles et al.²¹ Two studies had a cross-over design^{16,17} and one a controlled reversal design.⁶ Three studies included a pre/post-design^{18–20} and one a cross-sectional case-control design.²² A within-group comparison was used in six studies; in the remaining two, a between-group comparison was used.^{21,22} The entire study-population included 219 persons (range 10–100 persons), where 158 were transtibial amputees and 61 were transfemoral amputees. Mean age was 48.5 years with a range from 18–83 years.

Type of intervention and control intervention

Seven of the eight studies included a new prosthesis intervention compared to another prosthesis intervention,

which, in most cases, was the person's 'old' prosthesis system (Table 1). One study included a comparison of two patient groups already fitted with two different types of prosthesis.²² The most common type of intervention was the MP-knee, which was compared to a mechanical controlled knee – a NMP-knee.^{17,18,20,22} Klute et al.¹⁷ also included a comparison between a shock-absorbing pylon prosthesis compared to rigid pylon prosthesis. The remaining four studies concerned different prosthesis interventions: a silicon cover prosthesis compared to a non-silicon prosthesis¹⁹; an OI-prosthesis compared to a conventional socket prosthesis⁶; an elastomeric suspension liner with distal locking pin compared to a liner with neoprene suspension sleeve¹⁶; and a total surface-bearing socket compared to a conventional patellar tendon-bearing socket.²¹

Outcome evaluation instruments

The included studies reported results from 14 different instruments for outcome evaluation (Table 2). A majority of the instruments focused on activity ($n = 8$) and/or QoL ($n = 8$). Some of the instruments targeted user satisfaction and prosthesis use, and one instrument described utility in relation to costs. Another instrument was about risk factors and some instruments included questions from more than one area.

Activity and participation outcomes

Five studies reported significant improvements in activity and/or participation (Table 3).^{6,16,18,19,20} Neoprene suspension sleeve increased time spent on activities/day.¹⁶ Using prosthesis with silicone cover showed an increase in engagement in predefined everyday activities involving revealing the body.¹⁹ Outdoor mobility was found to improve with the OI-prosthesis⁶ and Hafner et al.²⁰ found that activities that require divided attention such as stair and hill descents went faster with the MP controlled knee-prosthesis compared to the NMP prosthesis. Hafner et al.²⁰ as well as Kahle et al.¹⁸ found that the frequency of stumbles and falls decreased with the MP-knee. In two studies, the authors reported no effects from intervention on activity and/or participation. Selles et al.²¹ compared a total surface-bearing socket with a patellar tendon-bearing socket and found no difference in mobility-related activities. Klute et al.¹⁷ could not demonstrate any difference in level or duration of activity when comparing a shock absorbing versus rigid pylons prosthesis or an MP-knee versus a NMP-knee.

Quality of life outcomes

An increase in aspects related to QoL was reported in four studies (Table 3).^{6,18,20,22} The intervention in one of these studies was the OI prosthesis⁶ and in the other three studies the MP knee. No difference in QoL was reported in the study by Hafner et al.²⁰ except for one issue – frustration with falling – for the benefit of the MP-knee. Donovan-Hall

Table 4. Summary of internal validity assessment of included studies (modified from Borghouts et al. 1998).*

Reference	Selection of population described	Inclusion and exclusion criteria described	Study size > 10 patient years	Follow up > 4 months	Dropouts <20%	Description of dropouts	Outcome measures congruent with aims	Data presentation congruent with aims	Confounders adjusted in the analysis	Psychometric properties of instruments reported	Total score
Coleman et al. 2004 ¹⁶	0	1	0	0	1	1	1	1	0	0	5
Donovan-Hall et al. 2002 ¹⁹	1	1	0	0	1	0	1	1	0	1	6
Gerzeli et al. 2009 ²²	1	1	1	NA	1	1	1	1	0	1	8
Hafner et al. 2007 ²⁰	1	1	0	0	1	1	1	1	1	0	7
Hagberg et al. 2008 ⁶	1	1	1	1	1	1	1	1	0	1	9
Kahle et al. 2008 ¹⁸	0	1	0	0	1	1	1	1	0	1	6
Klute et al. 2006 ¹⁷	1	1	0	0	0	1	1	1	0	1	6
Selles et al. 2005 ²¹	1	1	0	0	0	1	1	1	0	0	5

*If the study fulfilled the criteria, it was assigned '1'; if not '0'. NA, Not applicable.

et al.¹⁹ reported no change in terms of self-esteem, anxiety or depression when using a silicon-covered prosthesis compared to non-silicon prosthesis.

User satisfaction, use/non-use, and costs

Four studies reported results in terms of user satisfaction.^{16,18,20,21} Two of those studies^{18,20} reported on satisfaction with the MP-knee versus the NMP-knee. In both studies, a majority of subjects were more satisfied with the MP-knee. Selles et al.²¹ reported no difference in user satisfaction with a total surface-bearing socket compared to a conventional. Coleman et al.¹⁶ reported no difference in user satisfaction with the Alpha[®] versus the Pe-LiteTM liner.

Hagberg et al.,⁶ Coleman et al.¹⁶ and Hafner et al.²⁰ all reported on results in terms of use/non-use. An increased use of the OI-prosthesis system compared to the 'old system' was reported by Hagberg et al.,⁶ and Hafner et al.²¹ demonstrated an increase use of the MP knee compared to the NMP-knee. Coleman et al.¹⁶ reported more time was spent wearing the Pe-Lite limb compared to the Alpha[®].

Gerzeli et al.²² and Selles et al.²¹ included information about costs/cost-effectiveness. Gerzeli et al.²² analysed cost-utility showing a gain in a five-year healthcare system perspective of 0.09 QALY (Quality Adjusted Life Years) for the MP-knee at a cost of €35.971 per QALY, which should be considered as cost effective. From a societal perspective with respect to costs, there was no difference between interventions in a five-year perspective. Selles et al.²¹ presented a comparison of costs for two kinds of sockets; the study reported on significantly higher cost for materials, a shorter manufacturing time, and fewer visits for the total surface bearing socket group compared to the conventional patellar tendon-bearing socket group.

Quality assessment

Internal validity. The evaluation of internal validity resulted in scores ranging from 5–9 (Table 4). The two studies with the highest scores had 8–9/10 'yes'.^{6,22} Inclusion and exclusion criteria were described in all studies. Outcome measures as well as data presentation were considered to be congruent with aims in all studies. Only two studies fulfilled the study size criterion (> 10 patient years),^{6,22} and one study fulfilled the criterion of follow-up time (> 4 months).⁶ The only RCT study²¹ fulfilled four of the seven quality assessment criteria (Table 5).

External validity and applicability. The external validity and applicability evaluation process resulted in scores ranging from 2–4 (Table 6). All studies were considered to have 'study participants described in detail' and 'clinically

Table 5. Quality assessment of controlled studies (modified from Van Tulder 2003).

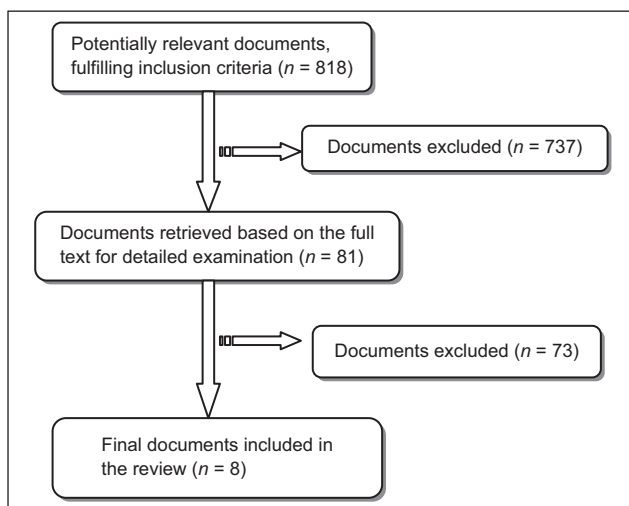
Reference	1	2	3	4	5	6	7
Selles et al. 2005 ²¹	Don't know	Don't know	Yes	Yes	Yes	Yes	No

1. Was the method of randomisation adequate? Yes/No/Don't know*, 2. Was the treatment allocation concealed?, 3. Were the groups similar at the baseline regarding the most important prognostic indicators?, 4. Were co-interventions avoided or similar?, 5. Was the compliance acceptable in all groups?, 6. Was the timing of the outcome assessment in all groups similar?, 7. Did the analysis include an intention-to-treat analysis?

Table 6. Summary of external validity and applicability assessment of included studies (modified from Shekelle et al. 1994).

Reference	Study participants described in detail	Intervention described in detail	Clinically relevant outcomes measured and reported	Size of the effect clinically important (at least 10% gain)	Total score
Coleman et al. 2004 ¹⁶					4
Donovan-Hall et al. 2002 ¹⁹					4
Gerzeli et al. 2009 ²²		0		0	2
Hafner et al. 2007 ²⁰				0	3
Hagberg et al. 2008 ⁶					4
Kahle et al. 2008 ¹⁸					4
Klute et al. 2006 ¹⁷				0	3
Selles et al. 2005 ²¹				0	3

1. Are the patients described in detail so that you can decide whether they are comparable to those that you see in your practice?, 2. Are the interventions and control interventions and treatment settings described sufficiently well to enable you to provide the same for your patients?, 3. Were outcomes measured and reported clinically relevant?, 4. Is the size of the effect clinically important (at least a 10% gain)?

**Figure 1.** Flowchart illustrating the literature selection process.

relevant outcomes measured and reported'. Four studies were considered to fulfil the criterion 'size of the effect clinically important (at least 10% gain)',^{6,16,18,19} and only one study did not fulfil the criteria 'intervention described in detail'.²²

Summary of results

Types of prosthesis interventions as well as assessment tools varied widely in included studies. For example, four of the studies were unique in being the only study reporting on a particular kind of intervention and consequently it was not possible to summarise any outcome from those interventions (Table 1 and 3). Still, four studies compared the MP-knee and the NMP-knee.^{17,18,20,22} Regarding activity, participation and user satisfaction, the results of these studies were inconsistent; however, in one study the MP-knee was found to decrease stumbles and falls¹⁸ and in another study the MP-knee was found to improve the patient's ability to descend stairs and hills.²⁰ Klute et al.¹⁷ reported no difference in terms of daily activity level or duration in transfemoral amputees with MP knee versus NMP knee. As to user satisfaction, the MP-knee was rated higher compared to the NMP-knee in two studies.^{18,20} All three studies that reported results in terms of QoL presented improvement with the MP-knee.^{18,20,22} Hafner et al.²⁰ reported that the MP-knee was used more often than the NMP-knee. Gerzeli et al.²² included an analysis of cost-utility. From a healthcare perspective, the MP-knee was considered to be

cost effective, expressed in terms of QALY. When non-healthcare costs and productivity losses were included, costs did not differ between interventions.

The quality of the included studies varied. The internal validity ranged from 5 to 9 (max 10), and only the study by Hagberg et al.⁶ fulfilled the criterion of follow-up time (> 4 months). However, inclusion and exclusion criteria were described in all studies. The external validity and applicability ranged from 2 to 4 (max 4). Four studies were considered to fulfil the criterion 'size of the effect clinically important, at least 10% gain'.^{6,16,18,19}

Discussion

Geil wrote; 'Evidence-based practice combines a practitioner's training and experience with evidence established through scientific research'.²³ This systematic review identifies research presented in scientific literature on effectiveness of lower-limb prosthesis and assesses the quality of the identified research. The target-group for the review was adults and the effects from prosthetic interventions was to be presented as change in activity, participation, and/or QoL as primary outcome variables, and user satisfaction, use/non-use in real life contexts, and cost-effectiveness as secondary outcomes. Results from a well-conducted review should help develop clinical recommendations based on evidence on effectiveness, i.e. outcomes in real life context.⁹ Contrary to what we had hoped, we could not, in light of the findings presented, provide any clear recommendations based on available evidence. The small number of studies, varying quality and the variety of prosthetic systems presented, together with a diversity of outcome instruments, make clinical recommendations based on those results unreliable. Although some of the included studies were considered to have high internal and external validity, there were obvious shortcomings in several studies as regards to study size, which only exceeded 26 participants in one study, and follow-up time, which was less than four months in all but one study (Tables 4–5). The RCT study by Selles et al.²¹ was found to have shortcomings in one vital aspect – method of randomization (Table 5).

The selection process is a critical phase in a review. To increase the reliability and provide valid results, two researchers independently scanned the search results and selected studies for inclusion and exclusion. The authors of this study are all rehabilitation professionals, but with only little experience in the prosthetic field. This could be considered a weakness, since there was little pre-understanding and prior knowledge in the field of prosthetics and consequently no prior estimations on the scope of findings. However, all authors are experienced in studying outcome from other kinds of assistive technology and in doing reviews in this field.^{11,12} Internal and external validity as well as applicability and quality assessment were discussed

in the group of researchers until consensus was reached. The included methods for assessment have all been used in two former reviews conducted by the same Nordic therapists' research team.^{11,12} The overall assessment process was thus well established.

We are fully aware that our review presents significant challenges because it focuses on how a specific intervention such as a lower-limb prosthesis intervention could affect the daily life of its users (*effectiveness*) and excludes studies that report effects from data collected in a standardised environment or activity (*efficacy*). However, it is very important to gain more knowledge about how an assistive device, such as prosthesis, affects patients who have left hospital care and are living in their natural contexts. Based on this review, it seems to be a great challenge to perform well-designed studies that can deliver evidence in terms of effectiveness, in the AT field.

By evaluating existing research in a systematic and explicit way, clinical guidelines and recommendations for future interventions could be based on judgements with reduced errors. In addition, evidence-based guidelines could help patients acknowledge realistic expectations with respect to specific interventions. This type of realistic expectation has been found to affect the final outcome positively.²⁴

Although the included studies had relatively high levels of internal and external validity, the true overview of findings in terms of activity, participation and QoL in real-life contexts was inconclusive and vague and consequently the described interventions could not be recommended based on evidence. Just referring to the design of the included studies (Table 1), one could state that the quality of evidence as it is generally defined (for example, by the Grade Working Group, www.GradeWorking-Group.org) can be considered as low in the referenced studies. The question is then this: is it possible to implement studies with a high level of evidence within the AT-field while still uncovering results related to aspects of daily life? Other authors of systematic literature reviews in this field have made similar conclusions and have also highlighted the need for further research.^{25–27} Thus, effects from AT, such as the lower limb prosthesis, present a major challenge to standard evidence grading methods.²⁸

Conclusion

Seven studies reported significant improvements in one or more of the identified outcomes. However, only two of these studies obtained a high score in both internal and external validity. Interventions and outcome measurement methods varied considerably between studies; generally there were also few participants and short follow-up time. Research related to which key factors affecting the effectiveness of prosthetic interventions on user's everyday life

still remains to be done and should be of high interest and priority to clinicians in this field.

Acknowledgements

We would like to thank the librarians at Stakes (now National Institute for Health and Welfare) (Helsinki, Finland) for their substantial help with the electronic searches.

Funding

This research was supported in part by Nordic Development Centre for Rehabilitation Technology (now Nordic Welfare Centre).

References

1. Fergason JR and Boone DA. Custom design in lower limb prosthesis for athletic activity. *Phys Med & Rehabil Clin North Am* 2000; 11(3): 681–99.
2. Carroll K. Adaptive prosthetics for the lower extremity. *Orthot Prosthet* 2001; 6(2): 371–86.
3. Scherman RA. Utilization of prosthesis among US veterans with traumatic amputation: A pilot survey. *J Rehabil Res Dev* 1999; 36(2): 100–08.
4. Poljak-Guberina R, Zivkovic O, Muljagic A, Guberina M and Bernt-Zivkovic T. The amputees and quality of life. *Coll Antropol* 2005; 29(2): 603–09.
5. Karmarker AM, Collins DM, Wichman T, Franklin A, Fitzgerald SG, Dicianno BE, Pasquina PF and Cooper RA. Prosthesis and wheelchair use in veterans with lower-limb amputation. *JRRD* 2009; 46(5): 567–76.
6. Hagberg KBR, Guntherberg B and Rydevik B. Osseointegrated trans-femoral amputation prostheses: Prospective results of general and condition-specific quality of life in patients at 2-year follow-up. *Prosthet Orthot Int* 2008; 32(1): 29–41.
7. Sullivan J. UM, Robinson K and Sooriakumaran S. Rehabilitation of the trans-femoral amputee with an osseointegrated prosthesis: the United Kingdom experience. *Prosthet Orthot Int* 2003; 27: 114–20.
8. Van der Linde, Hofstad CJ, Geurts ACH, Postema K, Geertzen JHB and van Limbeek J. A systematic review of the effect on different prosthetic components on human functioning with lower-limb prosthesis. *JRRD* 2004; 41(4): 555–70.
9. Gartlehner G, Hansen RA, Nissman D, Lohr KN and Carey TS. A simple and valid tool distinguished efficacy from effectiveness studies. *J Clin Epidemiol* 2006; 59(10): 1040–48.
10. International Organization for Standardization. *Assistive products for persons with disabilities – classification and terminology*. DS/EN ISO 9999 (E). Geneva: ISO Copyright Office, 2007.
11. Salminen A-L, Brandt Å, Samuelsson K, Töytäri O and Malmivaara A. Mobility devices to promote activity and participation: A systematic review. *J Rehabil Med* 2009; 41: 697–706.
12. Brandt Å, Samuelsson K, Töytäri O, Salminen A-L. Activity & participation, quality of life, and user satisfaction outcomes of environmental control systems and smart home technology: A systematic review. *J Disabil Rehabil Assistive Technol* 2010.
13. Borghouts A, Koes BW and Bouter LM. The clinical course and prognostic factors of non-specific neck pain: a systematic review. *Pain* 1998; 77: 1–13.
14. Shekelle PG, Andersson G, Bombardier C, Chertkin D, Deyo R, Keller R et al. A brief introduction to the critical reading of the clinical literature. *Spine* 1994; 19(Suppl 18): 2028–31.
15. Van Tulder M, Malmivaara A, Hayden J and Koes B. Statistical significance versus clinical importance trials on exercise therapy for chronic low back pain as example. *Spine* 2007; 32: 1785–90.
16. Coleman KL, Boone DA, Laing LS, Mathews DE and Smith DG. Quantification of prosthetic outcomes: Elastomeric gel liner with locking pin suspension versus polyethylene foam liner with neoprene sleeve suspension. *J Rehabil Res Dev* 2004; 41(4): 591–602.
17. Klute GK, Berge JS, Orendurff MS, Williams RM. Prosthetic Intervention effects on activity of lower-extremity amputees. *Arch Phys Med Rehabil* 2006; 87: 717–22.
18. Kahle JT, Highsmith MJ and Hubbard SL. Comparison of nonmicroprocessor knee mechanism versus C-Leg on Prosthesis Evaluation Questionnaire, stumbles, falls, walking tests, stair descent, and knee preference. *J Rehabil Res Dev* 2008; 45(1): 1–14.
19. Donovan-Hall MK, Yardley L and Watts RJ. Engagement in activities revealing the body and psychosocial adjustment in adults with a trans-tibial prosthesis. *Prosthet Orthot Int* 2002; 26: 15–22.
20. Hafner BJ, Willingham LL, Buell NC, Allyn KJ, Smith DG. Evaluation of function, performance, and preference as transfemoral amputees transition from mechanical to microprocessor control of prosthetic knee. *Arch Phys Med Rehabil* 2007; 88: 207–17.
21. Selles RW, Janssens PJ, Jongenengel CD and Bussmann JB. A randomized controlled trial comparing functional outcome and cost efficiency of a total surface-bearing socket versus a conventional patellar tendon-bearing socket in transtibial amputees. *Arch Phys Med Rehabil* 2005; 86: 154–61.
22. Gerzeli S, Torbica A and Fattore G. Cost utility analysis of knee prosthesis with complete microprocessor control (C-leg) compared with mechanical technology in trans-femoral amputees. *Eur J Health Econ* 2009; 10: 47–55.
23. Geil MD. Assessing the state of clinically applicable research for evidence-based practice in prosthetics and orthotics. *JRRD* 2009; 46(3): 305–14.
24. Judge A, Cooper C, Arden NK, Williams S, Hobbs N, Dixon D, Günther KP, Dreinhoefer K and Dieppe PA. Pre-operative **expectation** predicts 12-month post-operative **outcome** among patients undergoing primary total hip replacement in European orthopaedic centres. *Osteoarthritis and Cartilage*, Osteoarthritis Research Society 2011; 19(6): 659–67.

25. Hofstad C, Van der Linde H, Van Limbeek J and Postema K. Prescription of prosthetic ankle-foot mechanisms after lower limb amputation. *Cochrane Database Syst Rev* 2004; (1): CD003978.
26. Horgan O and MacLachlan M. Psychosocial adjustment to lower-limb amputation: A review. *Disabil Rehabil* 2004; 26(14/15): 837–50.
27. Geertzen JHB, Martina JD and Rietman HS. Lower limb amputation. Part 2: Rehabilitation – a 10 year literature review. *Prosthet Orthot Int* 2001; 25: 14–20.
28. Johnston M, Sherer M, Whyte J. Applying evidence standards to rehabilitation research. *Am J Phys Rehabil* 2006; 85: 292–309.